Evaluating LLMs as Verifiers for Scientific Property Estimates

A Technical Report

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Abstract

LLM-based discovery systems require not only generators, but also verifiers that can assess whether reported scientific results are plausible. This report evaluates current LLMs on a controlled verification task for materials-related property estimates. Using the principle-aware PiFlow pipeline as the discovery backbone, we construct pairs of candidate compounds and their physical property values across three scientific domains and ask ten LLMs to judge whether estimated physical property values are plausible under different prompting and sampling settings. We find that sampling temperature has limited aggregate impact, while prompt format substantially affects acceptance behavior: binary prompts lead to higher acceptance rates than ternary prompts. Stronger model variants are generally more accurate, but acceptance tendency and directional error bias remain model-family specific. Across models, rejected estimates are more often labeled as too low than too high, revealing a directional bias. These results suggest that LLM-only verification can provide useful signals, but remains insufficient as a standalone scientific verifier without calibration or external grounding.

1 Introduction

Recent LLM-based scientific agents increasingly combine hypothesis generation, experimental design, and result interpretation within a single discovery loop. In materials science and related domains, systems such as PiFlow formulate discovery as an information-seeking process guided by scientific principles [5]. A similar pattern appears in automated mathematics research, where agents generate candidate proofs and then route them through a verification stage before accepting a result [1]. These systems have made generation more accessible: a model can propose a candidate material, molecule, or theorem at low cost and at large scale.

However, current verification is less mature. Generated candidates are useful only if the system can distinguish a valid result from a plausible but incorrect one. This is particularly important for LLM-driven pipelines, where fluent explanations can mask weak domain reasoning or hallucinated details. Existing agentic systems therefore often separate the proposal step from an explicit verifier. For example, Aletheia uses a verifier subagent to filter model-generated mathematical results, and PiFlow's uncertainty-reduction loop depends on the reliability of the evidence returned for each hypothesis [1, 5]. In practice, a discovery system without a dependable verification stage risks converting generation errors into apparent scientific progress. Therefore, the validity of verification is crucial in scientific discovery.

This report studies a narrower and measurable version of that problem: whether current LLMs can judge reported scientific property estimates. In data-driven materials discovery, a candidate result often has the form of a proposed compound and an associated physical property value. We therefore ask models to classify a reported value as `accurate`, `too high`, or `too low` relative to a surrogate-predicted reference value. The benchmark is constructed by running PiFlow, perturbing the resulting metrics in a controlled way, and asking ten models from five families to judge the estimates. We focus on four sources of variation: sampling temperature (Section 3), prompt format (Section 4), model family and scale (Section 5), and directional error bias (Section 6).

2 Experimental Design

2.1 Discovery pipeline

We instantiate the principle-aware PiFlow pipeline [5] with `gemini-3.5-flash` as the discovery backbone and run it across PiFlow's three benchmark domains. Each domain pairs a search over a candidate space with a surrogate evaluator for the target property:

- **Nanohelix optimization** (`nanohelix`): a surrogate predicts the chirality g-factor of a nanohelix from its geometry; the physical ceiling is $g = 2$.
- **Molecular bioactivity** (`chembl`): a surrogate trained on the ChEMBL35 release of the ChEMBL bioactivity database [7, 2] predicts a molecule's bioactivity (pChEMBL value); the discovery target is 6.5.
- **Superconductor optimization** (`supercon`): a surrogate following Hamidieh [4] maps a chemical formula to its critical temperature T_c. The underlying data are the SuperCon-derived UCI superconductivity dataset [3, 6]; the target is the room-temperature goal of 298.15 K.

For each domain, the pipeline reasons from a scientific principle to a hypothesis, proposes an experimental candidate, and records the surrogate-predicted property value. The resulting candidates and predicted metrics define the reference set used in the judgment benchmark.

2.2 Constructing accurate and inaccurate estimates

To create a controlled verification benchmark, we treat each surrogate-predicted metric as the *accurate* reference value and generate two corrupted versions of it. The corrupted values are formed by adding or subtracting a fixed amount equal to 30% of the corresponding domain target and then clamping the result to the property's physical bounds (Table 1). Each candidate therefore contributes three reported values: unperturbed (`accurate`), inflated (`too high`), and deflated (`too low`). This construction gives an exactly balanced $\frac{1}{3}/\frac{1}{3}/\frac{1}{3}$ split of ground-truth labels. Because the perturbation is a fixed absolute shift rather than a percentage of the candidate's own value, some corrupted metrics are clipped at the boundary, such as a deflated T_c floored at 0 K or an inflated g-factor capped at 2.

Table 1: Metric perturbation per domain. The shift is 30% of the domain target, applied additively and then clamped to the physical bounds.

Domain	Property	Target	Physical bound	Shift (±)
`nanohelix`	chirality g-factor	2.0	[0, 2]	0.60
`chembl`	pChEMBL	6.5	[0, 6.5]	1.95
`supercon`	T_c (K)	298.15	[0, 298.15]	89.45

2.3 Judgment corpus and prompt formats

Each record contains a `question` (Scientific principles, rationale hypothesis, expected compound and a reported value), a `task type`, a `ground_truth` label, and the model's `answer`. The reported values are constructed so that exactly one third are accurate, one third too high, and one third too low. Three prompt formats are used: a *binary* form (`accurate` / `inaccurate`), a *ternary* form adding an explicit `unknown` option, and a *directional ternary* form (`accurate` / `too high` / `too low`). Table 2 summarizes the corpus.

Table 2: Corpus overview.

Records	5,400
Task types	chembl, nanohelix, supercon (1,800 each)
Models	5 families × {flagship, lite} = 10
Prompt groups	binary; ternary (+`unknown`); ternary (`too high`/`too low`)
Temperatures	directional task at $T = 0, 0.5, 1.0$; other groups at $T = 0$
Overall accuracy	64.7%

2.4 Alignment and metrics

Alignment. For temperature comparisons, each model receives the same 108 items in the same order. Likewise, all ten models within a given (prompt group, temperature) cell receive the same items. These comparisons are therefore item-paired. The three prompt groups use *different* reported values, so cross-prompt comparisons are distributional (paired by model, never by item).

Key metrics. Two quantities recur. The **accept rate** is $P(\text{answer} = \texttt{accurate})$: the fraction of estimates a model labels as accurate. Higher accept rates indicate a lower threshold for accepting reported values; lower accept rates indicate a more conservative threshold. The ground-truth accept rate is 33%. The **bias index** measures directional preference,

$$\text{bias} \;=\; \frac{N_{\texttt{too high}} - N_{\texttt{too low}}}{N_{\texttt{too high}} + N_{\texttt{too low}}} \;\in\; [-1, +1], \tag{1}$$

where positive over-calls `too high` and negative over-calls `too low`. Because ground truth is an even split, any deviation from 0 is model bias.

3 Effect of Temperature on Judgment

Aggregate behavior is nearly unchanged across temperature. On the directional task, accuracy and answer mix vary only slightly from $T = 0$ to $T = 1.0$ (Table 3). A Friedman test across the ten models finds no significant temperature effect on accuracy ($\chi^2 = 5.35$, $p = 0.069$) or on accept rate ($\chi^2 = 1.45$, $p = 0.48$), and no individual model shows a significant temperature $\times$ answer association under a per-model χ^2 test (all $p > 0.05$).

Table 3: Aggregate behavior by temperature (10-model mean), directional task.

Temperature	Accuracy	`accurate`	`too high`	`too low`
0.0	65.0%	24.4%	30.0%	45.5%
0.5	65.8%	25.1%	30.3%	44.3%
1.0	63.4%	25.8%	29.5%	44.7%

Item-level stability varies by model. The stable aggregate distribution hides a moderate amount of per-item variation: about 14% of items receive a different answer between $T = 0$ and $T = 1.0$. These changes largely cancel when averaged, but their frequency is model-specific. `gpt-5.4-mini` changes 32% of item-level answers between $T = 0$ and $T = 1.0$ and 53% across the three temperatures, whereas `gemini-3.1-flash-lite` changes 4% and `gpt-5.5` changes 8% (Figure 1b).

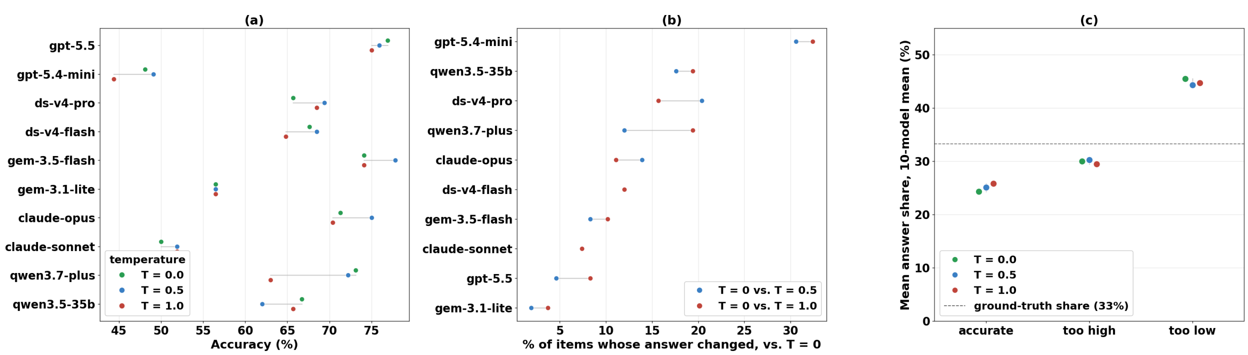

Figure 1: Visualization of the relationship between temperature and model judgements. (a) Per-model accuracy range across $T \in \{0, 0.5, 1.0\}$; dots are temperature-coloured and connected by their min–max span, showing that accuracy is nearly flat across temperatures for all models. (b) Per-item answer-flip rate (fraction of items whose label changed relative to $T = 0$), sorted by overall instability; the spread reveals a $\sim$14$\times$ range across models. (c) Aggregate answer-share for each label category (10-model mean); tightly clustered dots confirm that the collective answer mix barely shifts with temperature, while the dashed line marks the 33% ground-truth share.

Implication. In this setting, temperature is not a reliable way to control the distribution of judgments. It mainly affects repeatability at the item level, so reproducible labeling should use $T = 0$ and a model with low observed instability.

4 Effect of Prompt Format on Acceptance

Prompt format substantially changes how often models accept an estimate (Table 4, Figure 2). With only two choices, `accurate`/`inaccurate`, the mean accept rate is 38.3%, above the 33% truth share. When a third option is available, either `unknown` or an explicit `too high`/`too low` distinction, the accept rate falls to about 25%. The two ternary formats are statistically indistinguishable.

Table 4: Accept behavior by prompt format at $T = 0$ (10-model mean). "Recall" is the rate of saying `accurate` when the estimate truly is accurate; "False accept" is saying `accurate` when it is actually wrong.

Prompt format	Accept rate	Recall (true `accurate`)	False accept
binary (`accurate`/`inaccurate`)	**38.3%**	65.6%	24.6%
ternary (+`unknown`)	25.9%	48.6%	14.6%
ternary (`too high`/`too low`)	24.4%	45.6%	13.8%

A Friedman test across models is significant ($\chi^2 = 7.8$, $p = 0.020$). Pairwise Wilcoxon tests, paired by model, show a difference between the binary and directional-ternary prompts ($p = 0.002$, median model shift +12.9 pts) and between the binary and unknown-ternary prompts ($p = 0.035$, shift +17.1 pts). The two ternary formats do not differ ($p = 0.77$). When `unknown` is available, models use it in 25.5% of cases. The binary prompt increases both recall on truly accurate estimates (65.6% vs. 45.6%) and false acceptance of inaccurate estimates (24.6% vs. 13.8%), indicating a shift in acceptance threshold rather than improved discrimination.

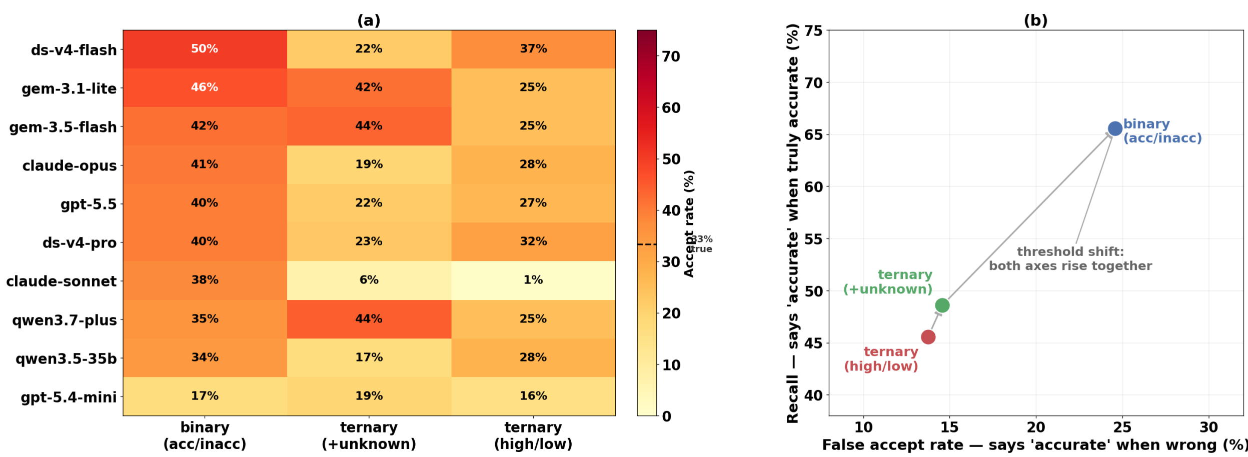

Figure 2: Effect of prompt format on model acceptance behaviour. (a) Accept rate P(*answer* = "accurate") for each model under three prompt formats (binary, ternary with *unknown*, ternary *too-high or too-low*); cells are sorted by binary accept rate and the colorbar reference marks the 33% true-accurate share. (b) ROC-style decomposition of accepts (10-model mean): false accept rate versus recall for each format; the diagonal arrow indicates a threshold shift — both axes rise together as the prompt becomes more permissive, with no improvement in discrimination.

Interpretation. Prompt design is the largest observed source of variation in acceptance rate. The binary format shifts models toward accepting more estimates, while both ternary formats make them more conservative. Accuracy should not be compared directly across prompt formats because the label spaces and chance levels differ.

5 Model Family and Scale

The stronger variant is more accurate in families involved in the experiment. Across the five families, the flagship or stronger variant outperforms the corresponding lite variant in all five pairs (mean 71.2% vs. 58.2%; Table 5, Figure 3a). With only five paired comparisons, a sign or Wilcoxon test cannot fall below $p = 0.0625$, so the result should be read as a consistent directional pattern rather than a powered statistical claim. The most accurate model overall is `gemini-3.5-flash` (77.8%); the least accurate is `claude-sonnet-4-6` (50.9%).

Table 5: Flagship vs. lite by family. Accuracy is over all conditions; accept rate is on the directional task at $T = 0$.

Family	Flag. acc	Lite acc	Gap	Flag. accept	Lite accept
gpt	74.4%	52.0%	+**22.4**	26.9%	15.7%
claude	68.1%	50.9%	+**17.2**	27.8%	0.9%
gemini	77.8%	62.2%	+**15.6**	25.0%	25.0%
qwen	68.7%	63.3%	+5.4	25.0%	27.8%
deepseek	67.0%	62.4%	+4.6	32.4%	37.0%

Acceptance tendency does not follow the same ordering. Unlike accuracy, the tendency to label an estimate as `accurate` is not consistently higher or lower for flagship models (Figure 3b). The `gpt` flagship accepts more often than its mini variant, whereas the `deepseek` and `qwen` lite variants accept more often than their stronger variants; the two `gemini` variants have the same accept rate. The clearest outlier is `claude-sonnet-4-6`, with an accept rate of only 0.9%. This near-zero acceptance rate causes it to miss many truly accurate estimates and contributes to its low overall accuracy. The Claude family is also the most conservative family and the only one with a family-level lean toward `too high` labels (Section 6).

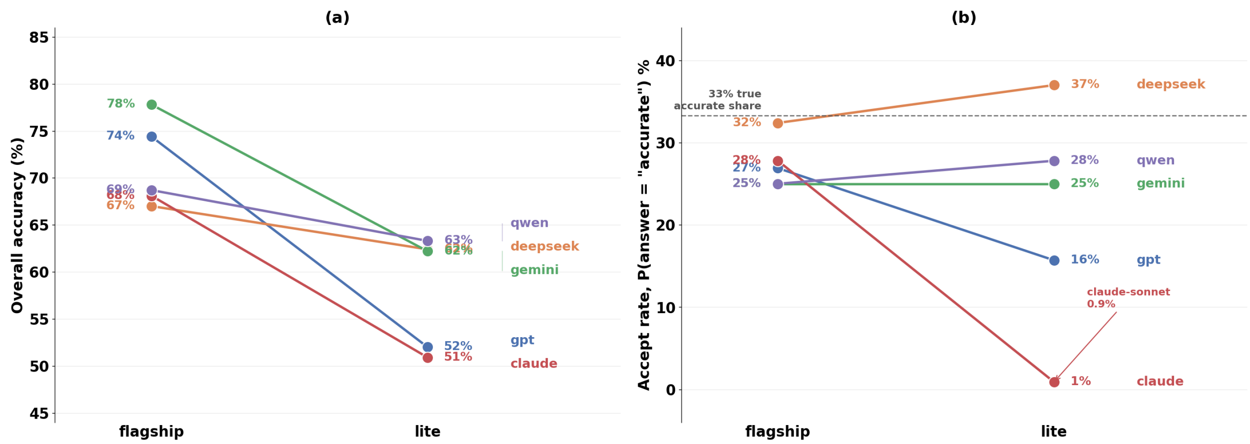

Figure 3: Flagship versus lite model performance by family. (a) Overall accuracy for the flagship and lite variant of each of the five model families; lines descend in all cases, confirming a consistent scale advantage. (b) Accept rate under the directional prompt at T=0; the dashed line marks the 33% true-accurate baseline. Claude-sonnet is an outlier with a near-zero accept rate (0.9%), while DeepSeek-lite exceeds its flagship counterpart.

Implication. Larger or stronger models provide a clear accuracy benefit in this benchmark, but they do not remove prompt sensitivity or family-specific acceptance behavior. Accuracy and calibration therefore need to be evaluated separately.

6 Directional Error Bias

The model population has a slight lean toward `too low`. Although the ground truth is balanced, the ten models collectively output `too high` 969 times and `too low` 1,453 times. This corresponds to a bias index of -0.20 and is significant under a binomial test ($p \approx 7 \times 10^{-23}$). The bias is stable across temperature, with indices of -0.21, -0.19, and -0.21 at $T = 0$, $T = 0.5$, and $T = 1.0$, respectively.

Seven of ten models lean toward `too low`, and eight have a statistically significant directional bias (Table 6, Figure 4). The largest negative bias indices occur for `gemini-3.1-flash-lite` (-0.47), `gpt-5.4-mini` (-0.45), and `qwen3.7-plus` (-0.41). The only models with a positive bias are the two Claude models and `gemini-3.5-flash`; among them, only `claude-opus-4-8` is significant ($+0.22$). Consistent with this asymmetry, models identify truly deflated estimates more often than truly inflated estimates: mean recall is 80.6% for true `too low` items and 68.2% for true `too high` items.

Table 6: Directional bias per model, pooled over temperatures. * denotes a significant deviation from 50/50 (binomial, $p < 0.05$).

Model	$N_{\texttt{too high}}$	$N_{\texttt{too low}}$	Bias index	Lean
`gpt-5.5`	85	146	−0.264*	`too low`
`gpt-5.4-mini`	72	191	−0.452*	`too low`
`deepseek-v4-pro`	80	131	−0.242*	`too low`
`deepseek-v4-flash`	71	123	−0.268*	`too low`
`gemini-3.5-flash`	136	111	+0.101	`too high`
`gemini-3.1-flash-lite`	64	179	−0.473*	`too low`
`claude-opus-4-8`	141	90	+0.221*	`too high`
`claude-sonnet-4-6`	171	150	+0.065	`too high`
`qwen3.7-plus`	72	172	−0.410*	`too low`
`qwen3.5-35b-a3b`	77	160	−0.350*	`too low`

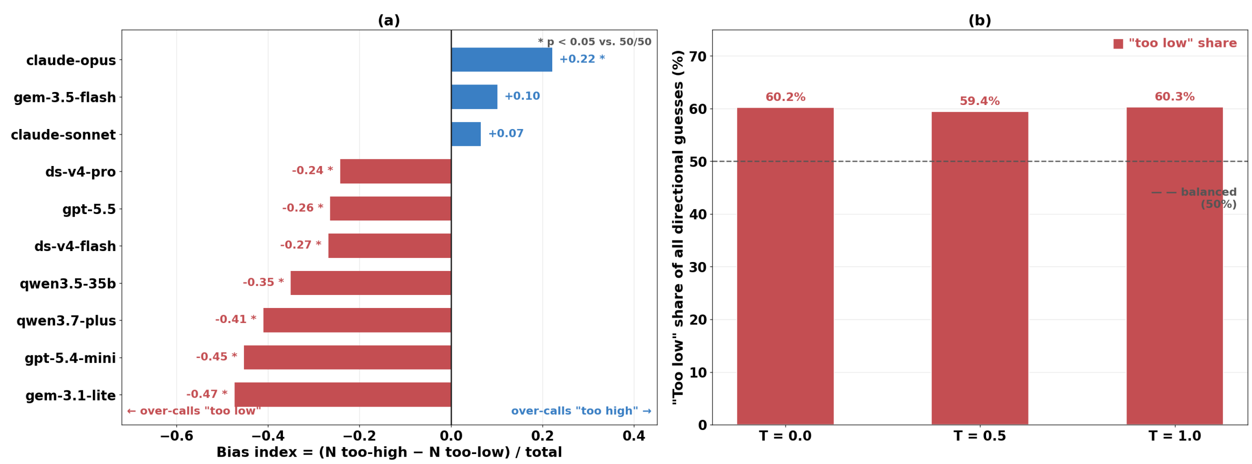

Figure 4: Directional error bias in model judgements. (a) Bias index = $(N_{\text{too-high}} - N_{\text{too-low}})/N_{\text{total}}$ pooled over all temperatures; negative values (red) indicate a systematic tendency to label estimates as "too low." Asterisks mark models whose deviation from a balanced 50/50 split is statistically significant ($p < 0.05$, binomial test). (b) Share of directional guesses labelled "too low" aggregated over all ten models at each temperature; the proportion remains stable at $\approx 60\%$ across $T \in \{0, 0.5, 1.0\}$, indicating that temperature does not resolve the bias.

Interpretation. Rejected estimates are not distributed symmetrically across error directions. For most models, the default rejection label is `too low`, meaning that the model more often treats the reported value as an underestimate than as an overestimate. Downstream systems that use these labels should therefore calibrate directional outputs rather than treating `too low` and `too high` as equally informative.

7 Conclusion

This report evaluates whether current LLMs can serve as verifiers for scientific property estimates in a controlled materials-discovery setting. The results show that LLM judgments contain useful signal: on the directional three-label task, models perform well above chance. However, this signal is not yet reliable enough for standalone verification. Model behavior is sensitive to prompt format, stronger models are generally more accurate but retain family-specific acceptance patterns, and rejected estimates exhibit a systematic directional bias toward being labeled as `too low`.

These findings indicate that current LLMs have limited ability to judge the validity of scientific discoveries on their own. Their verification behavior is easily affected by prompt design and by model-specific priors, making their judgments unstable and potentially biased. Therefore, LLM-based verification should not be treated as a final assessment of scientific correctness. Reliable verification of scientific discoveries still requires external validation through computation, domain data, experiments, or other task-specific evidence.

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